

# **Implementation and Evaluation of a Semi-supervised Federated Learning Algorithm**

by

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# 1 Introduction

## 1.1 Problem Description and Motivation

Federated Learning (FL) is a decentralized machine learning paradigm that allows multiple distributed clients to collaboratively train a shared global model without transferring their raw, privacy-sensitive data to a central server. While foundational algorithms like Federated Averaging (FedAvg) [3] successfully maintain data privacy, they heavily rely on an unrealistic assumption: that all participating clients possess large, fully annotated datasets.

In real-world applications, particularly those involving resource-constrained edge devices (such as mobile phones or local hospital servers), continuous data annotation is economically expensive, time-consuming, and labour-intensive. This creates a severe **labeled data scarcity** problem. Consequently, a typical edge device will only hold a very small fraction of labeled samples alongside a massive volume of raw, unlabeled data.

This scarcity introduces two major challenges in standard federated architectures. First, training a local model exclusively on a tiny subset of labeled data often results in severe overfitting and poor generalisation due to class distribution mismatches. Second, attempting to use the global model to assign labels to the local unlabeled data generally fails because of data heterogeneity across the network. In federated networks, client data is typically **non-IID (Non-Independent and Identically Distributed)**, meaning each device generates unique, heavily skewed data distributions that do not uniformly represent the global population. Conversely, **IID (Independent and Identically Distributed)** data would imply that every client has a statistically identical dataset, which is rarely the case in practical scenarios.

The primary motivation of this project is to address this labeled data scarcity by exploring Semi-Supervised Federated Learning (SSFL) techniques. Specifically, we focus on a realistic deployment scenario defined in the literature as “**Case 1**” [2], wherein every single client in the federated network holds both a small labeled dataset and an abundant unlabeled dataset. To leverage this untapped unlabeled data, we will implement and evaluate the FedLabel algorithm [1]. This approach intelligently mitigates non-IID challenges by adaptively selecting between the locally fine-tuned model and the globally aggregated model to assign pseudo-labels to unlabeled data based on dynamic confidence metrics.

Furthermore, a significant secondary problem in current SSFL research is the lack of standardised environments to test and compare these complex semi-supervised methods.

The transition from solving the algorithmic challenge of labeled data scarcity to developing a software tool is necessary because evaluating varying SSFL strategies requires a consistent baseline. Therefore, the extended motivation of this project is to design a modular benchmarking framework that standardises data partitioning and evaluation, allowing researchers to seamlessly plug in and compare custom pseudo-labeling strategies under identical federated conditions.

## ***1.2 Scope of the Project***

The scope of this project is strictly defined around the implementation, evaluation, and software architecture of a semi-supervised federated learning framework.

### **In-Scope Activities**

- **Target Data Scenario:** The project is exclusively bounded to the “Case 1” [2] federated learning scenario, meaning every simulated client will possess both a labeled and an unlabeled subset of data.
- **System Simulation:** Development of a fully simulated federated network using Python and the PyTorch deep learning framework. This includes both cross-device (100 clients) and cross-silo (10 clients) settings.
- **Data Heterogeneity:** The implementation will support and evaluate both IID and non-IID data distributions (achieved via Dirichlet distribution partitioning).
- **Algorithm Development:** Implementation of the baseline FedAvg pipeline [3] and the core components of the FedLabel algorithm [1], including confidence-based model selection and global-local consistency regularisation.
- **Software Architecture:** Design of an Object-Oriented, modular benchmarking framework with standardised data augmentation and evaluation metrics.

### **Out-of-Scope Activities**

- **Alternative Data Scenarios:** The project will not investigate scenarios where only the server has labeled data (Case 2), where only a subset of clients have labeled data (Case 3), or fully unsupervised federated learning (Case 4) [2].
- **Physical Deployment:** Deploying the framework onto actual physical distributed

edge devices or real-world network infrastructure is outside the project boundaries; all operations will be simulated on centralised hardware.

- **Adversarial Defences:** Evaluating or implementing defences against advanced federated privacy attacks (e.g., gradient inversion, membership inference) is not included in the primary scope.

### 1.3 Definitions, Acronyms, and Abbreviations

The following terms and acronyms are used throughout this document:

| Term                                                         | Definition                                                                                                                                                                                       |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Case 1</b>                                                | A specific deployment scenario in Federated Learning with Unlabeled Data where every participating client possesses both a labeled data subset and an unlabeled data subset during training [2]. |
| <b>FedAvg (Federated Averaging)</b>                          | The foundational algorithm for federated learning that computes a global model by taking the weighted average of locally trained client models [3].                                              |
| <b>FL (Federated Learning)</b>                               | A machine learning paradigm where multiple clients collaboratively train a model without sharing raw local data.                                                                                 |
| <b>FLUD (Federated Learning with Unlabeled Data)</b>         | An umbrella term for research focusing on the integration of unlabeled data into federated learning pipelines [2].                                                                               |
| <b>IID (Independent and Identically Distributed)</b>         | A statistical property where data samples are drawn from the same probability distribution and are mutually independent. In FL, this implies all clients have representative, balanced datasets. |
| <b>Non-IID (Non-Independent and Identically Distributed)</b> | A state of data heterogeneity where client datasets differ significantly in size, class balance, and underlying distribution.                                                                    |
| <b>SSFL (Semi-Supervised Federated Learning)</b>             | A branch of machine learning that combines federated network architectures with semi-supervised algorithms to utilise both labeled and unlabeled data.                                           |

## 2 Related Work

### 2.1 *Background Concepts*

To establish a robust foundation for Semi-Supervised Federated Learning (SSFL), it is essential to delineate the convergence of decentralised training and semi-supervised optimisation strategies.

#### 2.1.1 Federated Learning Foundations

Federated Learning (FL) primarily addresses data privacy paradigms by enabling decentralised client nodes to train local models on their native storage, subsequently transmitting only model weights or gradients to a central orchestrating server. The standard baseline framework, Federated Averaging (FedAvg) [3], aggregates these local updates via a weighted average proportional to each client's dataset size. However, conventional FL assumes fully supervised environments, where every local data point is pre-annotated with ground-truth labels. In real-world edge execution environments, this assumption introduces significant operational bottlenecks due to the high fiscal and temporal costs of data labeling.

#### 2.1.2 Semi-Supervised Learning Techniques

Semi-Supervised Learning (SSL) mitigates the lack of annotated data by combining a limited corpus of labeled data with a vastly superior volume of unlabeled data. Modern centralised SSL frameworks rely heavily on two mechanisms:

- **Consistency Regularisation:** Enforcing the model to yield identical prediction distributions when exposed to different stochastic augmentations (weak vs. strong) of the same unlabeled input sample.
- **Pseudo-Labeling:** Utilising the model's highly confident predictions on unlabeled data as artificial ground-truth targets for subsequent supervised optimisation steps.

When brought into the federated domain, integrating SSL mechanisms introduces severe challenges, particularly because data distributions across independent client nodes are typically Non-Independent and Identically Distributed (non-IID), causing local models to diverge and generate unreliable pseudo-labels.

## 2.2 Analysis of State-of-the-Art Methods

Multiple paradigms have been introduced in recent literature to handle the challenges of Federated Learning with Unlabeled Data (FLUD), specifically within the context of “Case 1,” [2] where clients contain both labeled and unlabeled partitions.

### 2.2.1 Federated Averaging (FedAvg) Baseline Application

**Aim:** To establish a foundational distributed model aggregation mechanism without data centralisation.

**Methodology:** Local clients perform vanilla Stochastic Gradient Descent (SGD) utilising only their limited supervised data partition. The server aggregates parameters periodically based on data volume weighting [3].

**Experimental Setting:** Typically evaluated on standardised computer vision datasets (e.g., CIFAR-10, MNIST) distributed across 10 to 100 client configurations under varying Dirichlet distribution factors ( $\alpha$ ) to simulate non-IID conditions.

**Experimental Results:** Demonstrates stable convergence when local datasets are identically distributed (IID). However, accuracy degrades dramatically (often exceeding a 30% performance drop) under highly heterogeneous non-IID settings when labeled data is scarce.

**Strengths:** Extremely simple to implement, introduces low computational overhead on edge nodes, and requires minimal communication bandwidth.

**Weaknesses:** It completely ignores the vast volume of local unlabeled data available on client devices, leading to catastrophic overfitting on the micro-sized labeled subsets.

### 2.2.2 FedMatch (Jeong et al.)

**Aim:** To exploit unlabeled local data within a federated pipeline while explicitly combating client-side data heterogeneity [8].

**Methodology:** Introduces an inter-client consistency regularisation technique. Instead of relying solely on internal model consistency, FedMatch encourages a client’s local model to match the predictions of helper models transmitted from other clients with similar data distributions. It separates the learning of parameters into disjoint structures for labeled vs. unlabeled attributes.

**Experimental Setting:** Assessed using CIFAR-10 and SVHN datasets across a simulated network of 100 clients, evaluating both IID and complex non-IID partitions with very low annotation rates (e.g., only 10% labeled data per client).

**Experimental Results:** Outperforms baseline semi-supervised federated configurations, showing higher stability against non-IID class shifts and yielding noticeable classification accuracy improvements.

**Strengths:** Effectively reduces local model bias by verifying predictions against peer client architectures, resulting in higher-quality pseudo-labels.

**Weaknesses:** Introduces substantial communication overhead and privacy risks due to the explicit requirement of transmitting auxiliary parameters and tracking helper models across the distributed network.

### 2.2.3 RSCFed (Robust Semi-Supervised Federated Learning)

**Aim:** To harmonise global aggregation when facing highly unreliable local client models caused by severely restricted labeled samples.

**Methodology:** Employs a random sub-consensus aggregation mechanism on the server side. Instead of averaging all client updates universally, it sub-samples client clusters to perform localised consensus building, mitigating the negative impacts of highly divergent local pseudo-labels.

**Experimental Setting:** Evaluated over standard benchmark vision applications under severe label sparsity configurations, checking performance across extreme non-IID imbalances where certain clients completely lack specific target classes.

**Experimental Results:** Maintains steady model accuracy scaling even when over 90% of total client data points are completely unlabeled and data distributions are heavily skewed.

**Strengths:** Highly resilient to worst-case data heterogeneity and prevents low-quality client model updates from poisoning the global model parameters.

**Weaknesses:** The sub-consensus clustering strategy significantly extends the number of communication rounds required to achieve final model convergence, limiting its feasibility for real-time edge computing environments.

## 2.3 Synthesis and Structural Research Gaps

While existing frameworks like FedMatch [8] and RSCFed provide sophisticated workarounds for unlabeled data, they exhibit a major structural gap when executing under the precise conditions of Case 1 [2]. These methods either assume that a central server maintains a fully labeled proxy dataset to guide the network, or they inject immense communication strain by sharing model subsets between edge devices.

Furthermore, current frameworks fail to adaptively balance the knowledge inside the globally aggregated structure against the fine-tuned parameters inside the local client model when evaluating whether a pseudo-label is safe to use. This creates a clear research path for advanced frameworks like FedLabel [1] and FSSL-UC [5], which aim to dynamically control pseudo-label confidence using local-global consistency checking without increasing network transmission costs.

## 3 System Design

### 3.1 Flowchart and Pseudocode of Proposed Algorithms

The methodology relies on an advanced pseudo-labeling algorithm that leverages local-global model consensus to assign artificial labels to  $D_k^U$  securely. The full SSFL training loop for each communication round  $t$  proceeds through three main steps.

#### 3.1.1 Step 1: Obtaining the Global and Local Models

After receiving the global model  $W^t$  from the server, each client  $k$  initialises both its supervised local model and its semi-supervised model from the global model. The client then performs  $\tau$  SGD steps on its labeled dataset  $D_k^L$  to obtain the supervised local model  $w_{L,k}^{(t+\tau)}$  and computes the supervised local update  $\Delta w_{L,k}$ . This local model now contains the client's local knowledge and will be used as one of the two models for pseudo-labeling in the next step.

#### 3.1.2 Step 2: Confidence-Based Model Selection and Pseudo-Labeling

For each unlabeled sample  $\xi$  in the client's unlabeled set  $D_k^U$ , both the global model  $W^t$  and the trained local model  $w_{L,k}^{(t+\tau)}$  independently produce logit vectors  $s(W, \xi)$  and  $s(w_{L,k}, \xi)$ . A confidence score for each model's prediction is computed using a variance-

based function  $h(\cdot)$ , which quantifies the dispersion of the logit values; a higher variance signifies a more confident and peaked prediction.

Binary model selection then picks the logit vector with the higher confidence score as the pseudo-labeller for that sample:

$$s^*(\xi) = \arg \max \{h(s(W, \xi)), h(s(w_{L,k}, \xi))\}. \quad (1)$$

A hard pseudo-label  $\hat{y}_\xi$  is then assigned by taking the  $\arg \max$  of the selected logit vector  $s^*(\xi)$ , but only if the maximum predicted probability exceeds a confidence threshold  $\beta$ . Samples that do not meet the threshold are discarded from the unsupervised training batch, preventing noisy pseudo-labels from corrupting the model. The resulting pseudo-label and a strongly augmented view of the sample  $\hat{\xi}$  form the training pair for the cross-entropy component of the loss.

### 3.1.3 Step 3: Global-Local Consistency Regularization

The binary selection in Step 2 discards all information from the non-selected model's logits, denoted  $\bar{s}^*(\xi)$ . However, when both models agree on the same hard label for a sample (i.e., the  $\arg \max$  of the non-selected model's logits matches  $\hat{y}_\xi$ ), the soft output  $\bar{s}^*(\xi)$  still carries valuable distributional information that should not be ignored.

To exploit this complementary knowledge, a global-local consistency regularization term is introduced: a weighted KL-divergence between the training model's output distribution  $s(w_{U,k}, \xi)$  and the soft logits of the non-selected model  $\bar{s}^*(\xi)$ , applied exclusively to samples where both models share the same hard pseudo-label:

$$\lambda_k(\xi) \cdot \text{KL}(s(w_{U,k}, \xi) \parallel \bar{s}^*(\xi)) \cdot \mathbb{1}_{\{\arg \max_{i \in [N]} \bar{s}^*(\xi) = \hat{y}_\xi\}}. \quad (2)$$

The adaptive weight  $\lambda_k(\xi) = \lambda_0 \cdot h(\bar{s}^*(\xi)) / h(s^*(\xi))$  scales the regularization by the confidence ratio of the non-selected model to the selected model, with  $\lambda_0$  acting as a global scaling factor. This design ensures that the regularization is strongest when the two models are equally confident and weakest when the non-selected model is far less confident.

### 3.1.4 Final Semi-Supervised Loss

The complete semi-supervised objective for each client  $k$  at round  $t$  is obtained by averaging the per-sample contributions over the unlabeled dataset  $D_k^U$ :

$$F_{U,k}(w_{U,k}) = \frac{1}{|D_k^U|} \sum_{\xi \in D_k^U} \left[ \text{CE}(w_{U,k}, \hat{\xi}) + \lambda_k(\xi) \cdot \text{KL}(s(w_{U,k}, \xi) \parallel \bar{s}^*(\xi)) \cdot \mathbb{1}_{\left\{ \arg \max_i \bar{s}^*(\xi) = \hat{y}_\xi \right\}} \right]. \quad (3)$$

The client then performs  $\tau'$  SGD steps on  $D_k^U$  using this objective, initialised from the global model, to obtain the semi-supervised local update  $\Delta w_{U,k}$ . The final model update sent back to the server is the sum of both local updates:

$$\Delta w_k = \Delta w_{L,k} + \Delta w_{U,k}. \quad (4)$$

The server then aggregates these updates using weighted averaging:

$$W^{t+1} \leftarrow \sum_{k \in S_t} \frac{|D_k|}{\sum_{i \in S_t} |D_i|} W_k^t \quad (5)$$

completing the communication round.

---

#### Algorithm 1: SSFL Training – Server Orchestration

---

**Input:** Initial global model  $W^0$ ; total rounds  $T$ ; client set  $\mathcal{K}$

**Output:** Trained global model  $W^T$

**for** each communication round  $t = 0, 1, \dots, T - 1$  **do**

    Sample active client subset  $S_t \subseteq \mathcal{K}$

    Broadcast  $W^t$  to all clients  $k \in S_t$

**for** each client  $k \in S_t$  (in parallel) **do**

$\Delta w_k \leftarrow \text{CLIENTUPDATE}(W^t, k)$

// Algorithm 2

    // Weighted aggregation

$W^{t+1} \leftarrow \sum_{k \in S_t} \frac{|D_k|}{\sum_{i \in S_t} |D_i|} (W^t + \Delta w_k)$

**return**  $W^T$

---

---

**Algorithm 2: CLIENTUPDATE – Local Training at Client  $k$** 

---

**Input:** Global model  $W^t$ ; labeled set  $D_k^L$ ; unlabeled set  $D_k^U$ ; local steps  $\tau, \tau'$ ; threshold  $\beta$ ; scaling factor  $\lambda_0$

**Output:** Combined local update  $\Delta w_k$

// -- Step 1: Supervised local update --

$w_{L,k} \leftarrow W^t$

Perform  $\tau$  SGD steps on  $D_k^L$  starting from  $w_{L,k}$

$\Delta w_{L,k} \leftarrow w_{L,k}^{(t+\tau)} - W^t$

// -- Step 2 & 3: Semi-supervised local update --

$w_{U,k} \leftarrow W^t$  // initialise from global model

**for each unlabeled sample  $\xi \in D_k^U$  do**

    // Compute logits from both models

$s_G \leftarrow s(W^t, \xi)$

$s_L \leftarrow s(w_{L,k}^{(t+\tau)}, \xi)$

    // Binary model selection (Step 2)

**if**  $h(s_G) \geq h(s_L)$  **then**

$s^*(\xi) \leftarrow s_G, \bar{s}^*(\xi) \leftarrow s_L$

**else**

$s^*(\xi) \leftarrow s_L, \bar{s}^*(\xi) \leftarrow s_G$

    // Threshold-gated pseudo-label

**if**  $\max_i \text{softmax}(s^*(\xi))_i \geq \beta$  **then**

$\hat{y}_\xi \leftarrow \arg \max_i s^*(\xi)$

        Compute  $\text{CE}(w_{U,k}, \hat{\xi})$  // strong augmentation  $\hat{\xi}$

        // Global-local consistency regularization (Step 3)

**if**  $\arg \max_i \bar{s}^*(\xi) = \hat{y}_\xi$  **then**

$\lambda_k(\xi) \leftarrow \lambda_0 \cdot \frac{h(\bar{s}^*(\xi))}{h(s^*(\xi))}$

$\mathcal{L}(\xi) \leftarrow \text{CE}(w_{U,k}, \hat{\xi}) + \lambda_k(\xi) \cdot \text{KL}(s(w_{U,k}, \xi) \parallel \bar{s}^*(\xi))$

**else**

$\mathcal{L}(\xi) \leftarrow \text{CE}(w_{U,k}, \hat{\xi})$

**else**

        Discard  $\xi$  from training batch

$\mathcal{L}(\xi) \leftarrow 0$

// Minimise semi-supervised objective

$F_{U,k}(w_{U,k}) \leftarrow \frac{1}{|D_k^U|} \sum_{\xi \in D_k^U} \mathcal{L}(\xi)$

Perform  $\tau'$  SGD steps on  $D_k^U$  minimising  $F_{U,k}$

$\Delta w_{U,k} \leftarrow w_{U,k} - W^t$

// Combine and return

**return**  $\Delta w_k = \Delta w_{L,k} + \Delta w_{U,k}$

---

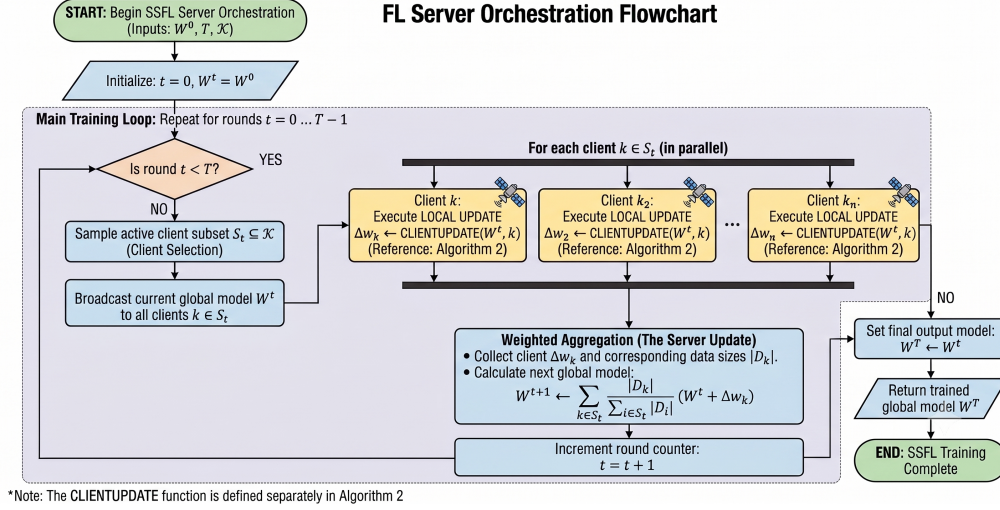


Figure 1: SSFL Training – Server Orchestration

### 3.2 Comparison Metrics

To rigorously evaluate the predictive performance and robustness of the proposed framework against State-of-the-Art (SOTA) baselines, we utilise a combination of standard machine learning classification metrics and relative performance ratios, aligning directly with our project specifications. Let  $TP$  denote True Positives,  $TN$  denote True Negatives,  $FP$  denote False Positives, and  $FN$  denote False Negatives.

- **Top-1 Classification Accuracy:** The fundamental ratio of correctly predicted observations to the total observations in the global test dataset. This serves as our primary absolute metric.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

- **Relative Accuracy Ratio (Method / FedAvg):** Because local pipeline variations make direct absolute comparisons with original papers unfeasible, scientific validity will be primarily assessed via relative performance gains. We will use the fully supervised FedAvg performance as a normalizing baseline to calculate the ratio:

$$\text{Relative Ratio} = \frac{\text{Accuracy}_{\text{FedLabel}}}{\text{Accuracy}_{\text{FedAvg}}} \quad (7)$$

- **Precision and Recall:** These intermediate metrics represent the ratio of correctly predicted positive observations to the total predicted positives (Precision), and the ratio of correctly predicted positive observations to all observations in the actual class

(Recall).

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

- **F1-Score:** The harmonic mean of Precision and Recall. Because non-IID federated distributions and specific medical imaging benchmarks lead to severe class imbalances, the F1-Score serves as a crucial secondary assessment criterion. It will be explicitly reported for the cross-silo experiments utilizing the medically imbalanced OrganAMNIST dataset [7].

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

### 3.3 Data Sets and Benchmarks

To benchmark the proposed system architecture, the experiments will be conducted utilising computer vision and biomedical datasets adapted for federated environments [2].

- **CIFAR-10 Dataset:** Utilised for cross-device experiments [6]. The labeled data ratio for these experiments will be set to 20% and 50%.
- **OrganAMNIST:** Utilised for cross-silo experiments [7]. To test more extreme label scarcity, the labeled ratio will be restricted to 5% and 20%.
- **PathMNIST:** Although not included in the original FedLabel study [1], it is incorporated into this project to further evaluate the algorithm’s capabilities [7].

#### Data Partitioning Strategy (Non-IID Simulation)

A centralised Data Distributor module will partition the full dataset across a configurable number of simulated clients. To simulate realistic data heterogeneity (non-IID conditions), data will be allocated using a Dirichlet distribution, denoted  $\text{Dir}(\alpha)$ .

- A small value (e.g.,  $\alpha = 0.1$ ) produces highly skewed, non-IID client datasets.
- A larger value (e.g.,  $\alpha = 1.0$ ) produces a more balanced, near-IID distribution.

Both configurations will be tested to evaluate the robustness of the framework. Each client’s local dataset is further partitioned into a small labeled subset and a large unlabeled subset using a uniform random split.

Furthermore, an augmentation factory will apply two levels of image transformations to the unlabeled data:

- **Weak augmentations:** Standard random horizontal flips and crops used for confidence-based model selection.
- **Strong augmentations:** Applied via RandAugment (magnitude=10, N=1) for the consistency loss, following the exact configuration of the reference paper [1].

## 4 System Architecture

The software architecture of the proposed Semi-Supervised Federated Learning (SSFL) framework is designed strictly around an Object-Oriented, modular three-tier structure. This decoupled approach ensures that data processing, local model optimisation, and global aggregation are logically separated, allowing independent modifications and seamless integration of alternative pseudo-labeling algorithms in the future.

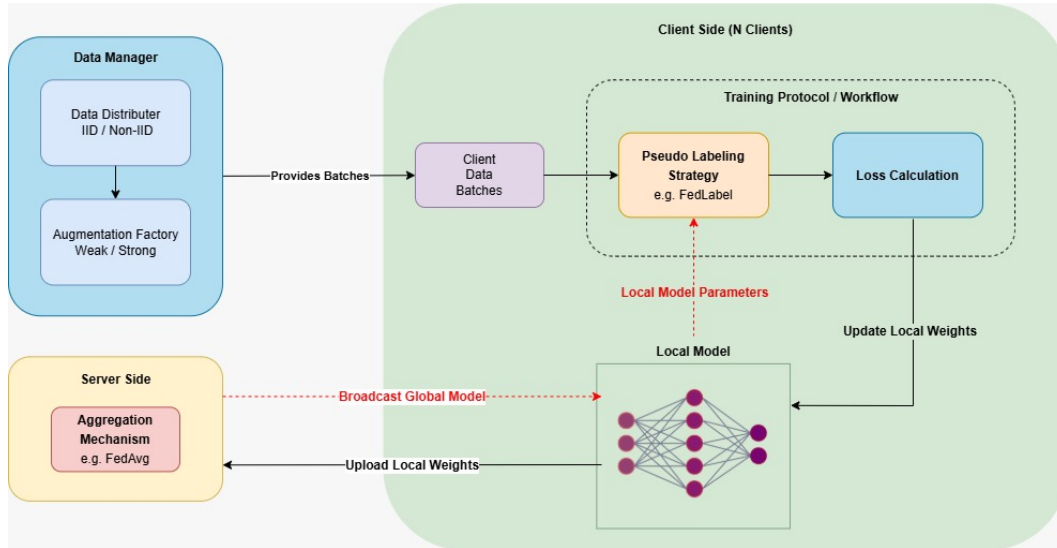


Figure 2: Proposed Modular System Architecture

### 4.1 Architectural Components

The system is divided into three primary functional modules.

## 1. Data Management Module

This module is solely responsible for data ingestion, preprocessing, and distribution prior to the commencement of the federated training loop.

- **Partitioner:** Implements the Dirichlet distribution mathematical logic to divide the global dataset into independent client subsets, effectively simulating realistic non-IID data skewness.
- **Augmentation Factory:** Generates multiple views of the same unlabeled input sample. It produces weakly augmented views (e.g., standard horizontal flips) for confidence evaluation and strongly augmented views (e.g., via RandAugment) for consistency regularisation loss computation.

## 2. Client-Side Execution Module

This component encapsulates the localised training logic executing independently on each simulated edge device.

- **Training Protocol Interface:** Manages the local epoch iterations and backpropagation mechanisms.
- **Pseudo-Labeling Engine:** Executes the dynamic variance-based confidence evaluation, deciding whether the local fine-tuned weights or the frozen global weights should dictate the artificial label for a given unlabeled sample.
- **Composite Loss Calculator:** Computes the combined gradient updates from the supervised cross-entropy loss and the global-local consistency KL-divergence loss.

## 3. Server-Side Aggregation Module

Acting as the central orchestrator, this module manages the global state of the neural network without ever accessing the clients' localised raw data. It handles client coordination, parameter reception, and the weighted averaging computations necessary to form the updated global model.

### 4.2 *Control Flow and Communication Protocol*

To explicitly define the interactions between the distributed components, the server-client communication protocol operates through a synchronised, iterative request-response cycle.

- **Client Selection (Orchestration Phase):** At the onset of communication round  $t$ , the Server Module executes a uniform random sampling function to select a specific fraction  $C$  of the total available clients. This simulates the real-world operational constraints of edge devices (e.g., battery life, network availability), ensuring that only a subset of devices actively participates in any given round.
- **Broadcast (Downlink):** The Server serialises the current global model parameters  $W^t$  and transmits them to the selected active client nodes.
- **Local Execution:** Each selected Client Module initialises its local architecture with the received  $W^t$ , processes its private labeled and unlabeled data batches, and calculates a localised weight update vector  $\Delta w_k$ .
- **Upload (Uplink):** The active clients transmit their localised weight updates  $\Delta w_k$  back to the Server Module. To ensure privacy, no raw data or direct label distributions are transmitted; only the numerical gradient updates cross the network boundary.
- **Global Aggregation:** Upon receiving the updates from all selected clients, the Server computes the weighted average of the parameters based on the dataset size of each contributing client, yielding the new global model  $W^{t+1}$  for the subsequent round.

## 5 Experimental Study

### 5.1 Experimental Setup

The framework will be benchmarked using a strictly simulated federated environment developed in Python using the PyTorch library. The experiments will be conducted under both cross-device and cross-silo configurations to comprehensively assess algorithmic robustness.

- **Network Configurations:** The cross-device setting will simulate 100 independent clients participating in the network, utilising the CIFAR-10 dataset [6]. The cross-silo setting will simulate 10 high-capacity clients utilising the OrganAMNIST and PathMNIST datasets [7].
- **Data Scarcity Ratios:** To enforce the “Case 1” [2] restrictions, the labeled data partition for the CIFAR-10 experiments will be artificially capped at 20% and 50% per

client. For OrganAMNIST and PathMNIST, more extreme label scarcity of 5% and 20% will be applied.

- **Heterogeneity Settings:** The Dirichlet distribution parameter  $\alpha$  will be toggled between  $\alpha = 1.0$  (representing mild, near-IID data distributions) and  $\alpha = 0.1$  (representing severe, highly skewed non-IID conditions).
- **Hyperparameter Isolation (Ablation):** To evaluate component impact, structured ablation studies will be conducted focusing on three core parameters: the maximum consistency weight scalar ( $\lambda_0$ ), the number of localised supervised epochs ( $\tau$ ), and the binary pseudo-label confidence threshold ( $\beta$ ).

## 5.2 Baseline Experimental Results

To establish a definitive lower bound for the proposed Semi-Supervised Federated Learning framework, we first evaluated the standard, fully supervised Federated Averaging (FedAvg) algorithm [3]. The models were trained over 1,000 communication rounds. The cross-device scenario (CIFAR-10) utilised a ResNet-34 architecture with a 10% client selection rate per round. The cross-silo scenarios (OrganAMNIST and PathMNIST) utilised a ResNet-18 architecture with a 100% client selection rate.

The results, detailed in Table 2, categorise the runs into “Low Baselines” (simulating the strict label scarcity of Case 1 [2]) and “High Baselines” (utilising 100% of available labels) across both severe non-IID ( $\alpha = 0.1$ ) and near-IID ( $\alpha = 1.0$ ) data distributions.

## 5.3 Discussions

The baseline experimental results clearly validate the core problem statement of this project. Under severe non-IID data partitions ( $\alpha = 0.1$ ), the FedAvg model overfits its restricted label sets, causing isolated predictions to become highly biased and resulting in accuracy degradation. For example, on the PathMNIST dataset under extreme non-IID conditions, increasing the label ratio from 20% to 100% resulted in a performance drop (from 28.08% to 21.65%), underscoring how conventional supervised aggregation fails when data is highly heterogeneous. The expected superiority of our proposed FedLabel framework [1] relies on resolving these exact vulnerabilities. By systematically forcing the client to defer to the globally aggregated model’s predictions when the local model’s confidence variance is low, the system is expected to bypass the local class-imbalance biases, thus producing significantly cleaner pseudo-labels and substantially improving

Table 2: FedAvg Baseline Experimental Results (1,000 Communication Rounds)

| Dataset & Architecture                                                | Baseline Type | Dirichlet ( $\alpha$ ) | Labeled Data Ratio | Accuracy (%) |
|-----------------------------------------------------------------------|---------------|------------------------|--------------------|--------------|
| <b>CIFAR-10</b><br>(ResNet-34, 100 Clients,<br>10% Selection Rate)    | Low Baseline  | 0.1                    | 20%                | 33.25        |
|                                                                       | Low Baseline  | 1.0                    | 20%                | 50.65        |
|                                                                       | Low Baseline  | 0.1                    | 50%                | 31.01        |
|                                                                       | Low Baseline  | 1.0                    | 50%                | 63.44        |
|                                                                       | High Baseline | 0.1                    | 100%               | 40.80        |
|                                                                       | High Baseline | 1.0                    | 100%               | 70.04        |
| <b>OrganAMNIST</b><br>(ResNet-18, 10 Clients,<br>100% Selection Rate) | Low Baseline  | 0.1                    | 5%                 | 49.25        |
|                                                                       | Low Baseline  | 1.0                    | 5%                 | 79.46        |
|                                                                       | Low Baseline  | 0.1                    | 20%                | 59.63        |
|                                                                       | Low Baseline  | 1.0                    | 20%                | 85.36        |
|                                                                       | High Baseline | 0.1                    | 100%               | 59.16        |
|                                                                       | High Baseline | 1.0                    | 100%               | 85.20        |
| <b>PathMNIST</b><br>(ResNet-18, 10 Clients,<br>100% Selection Rate)   | Low Baseline  | 0.1                    | 5%                 | 25.46        |
|                                                                       | Low Baseline  | 1.0                    | 5%                 | 61.95        |
|                                                                       | Low Baseline  | 0.1                    | 20%                | 28.08        |
|                                                                       | Low Baseline  | 1.0                    | 20%                | 76.50        |
|                                                                       | High Baseline | 0.1                    | 100%               | 21.65        |
|                                                                       | High Baseline | 1.0                    | 100%               | 71.81        |

upon the low baselines established in Table 2.

## 6 Project Status

### 6.1 Current State of the Project

As of the current phase of the engineering project, the foundational infrastructure of the simulation environment has been successfully completed. Preliminary experimental results evaluating the baseline FedAvg framework have also been completed and are reported in Section 5.2.

- **Literature and Scope Definition:** An exhaustive review of Semi-Supervised Federated Learning (SSFL) has been conducted, specifically delineating the operational boundaries of “Case 1” deployment scenarios [2].
- **Data Pipeline Execution:** The data management module is fully operational. The CIFAR-10 [6], OrganAMNIST, and PathMNIST datasets [7] have been successfully acquired and preprocessed. Furthermore, the Dirichlet distribution logic ( $\alpha = 0.1$  and

$\alpha = 1.0$ ) has been successfully implemented and tested to simulate realistic non-IID client heterogeneity.

- **Baseline Integration:** The standard supervised Federated Averaging (FedAvg) pipeline [3] has been coded, integrated, and actively tested.

During these initial implementations, the team encountered several technical difficulties, primarily involving computational bottlenecks due to hardware constraints, pipeline debugging, and ensuring the reproducibility of baseline metrics under severe data heterogeneity. These issues are currently being actively monitored and resolved through hyperparameter tuning and batch-size optimisation.

## 6.2 Task Log

Throughout the development cycle, continuous coordination has been maintained. Below is an excerpt from the formalised task log documenting critical project meetings.

### Meeting #1:

**Date:** 12/02/2026

**Location:** University Campus / Computer Engineering Department

**Period:** 13:00–14:30 (1.5 Hours)

**Attendees:** Assoc. Prof. Dr. Ömer Korçak, PhD Candidate Kübra Uludağ, Azra Çetintürk, Ozan Yıldırım

**Objectives:** Define the problem statement and review the FLUD taxonomy.

**Decisions and Notes:** The scope of the project was strictly bounded to the “Case 1” deployment scenario based on Kübra Uludağ’s review paper [2]. Initial literature review tasks were assigned.

### Meeting #2:

**Date:** 26/02/2026

**Location:** Online / MS Teams

**Period:** 14:00–15:30 (1.5 Hours)

**Attendees:** Assoc. Prof. Dr. Ömer Korçak, Azra Çetintürk, Ozan Yıldırım

**Objectives:** Finalise the algorithm selection for the SSFL framework.

**Decisions and Notes:** It was decided to proceed with the FedLabel algorithm [1] over FedMatch [8] due to its communication efficiency. The necessity of building a modular benchmarking framework was established.

### Meeting #3:

**Date:** 10/03/2026

**Location:** Library Study Room

**Period:** 15:00–17:00 (2.0 Hours)

**Attendees:** Azra Çetintürk, Ozan Yıldırım

**Objectives:** Internal team coordination for the dataset preparation phase.

**Decisions and Notes:** Ozan demonstrated the preliminary Dirichlet distribution logic for simulating non-IID conditions. Azra finalised the augmentation factory pipeline (RandAugment integration).

### Meeting #4:

**Date:** 02/04/2026

**Location:** University Campus / Computer Engineering Department

**Period:** 10:30–11:30 (1.0 Hour)

**Attendees:** Assoc. Prof. Dr. Ömer Korçak, Azra Çetintürk, Ozan Yıldırım

**Objectives:** Review the Object-Oriented software architecture proposed for the framework.

**Decisions and Notes:** The three-tier architecture (Data, Client, Server modules) was approved. The advisor emphasised making the pseudo-labeling engine an interchangeable interface for future integration of custom strategies.

### Meeting #5:

**Date:** 24/04/2026

**Location:** Online / Zoom

**Period:** 16:00–17:30 (1.5 Hours)

**Attendees:** PhD Candidate Kübra Uludağ, Azra Çetintürk, Ozan Yıldırım

**Objectives:** Resolve computational bottlenecks encountered during the FedAvg baseline execution.

**Decisions and Notes:** Consulted with Kübra Uludağ regarding Out-of-Memory (OOM) errors. Batch sizes were optimised, and PyTorch's `torch.autocast` was integrated to reduce memory consumption.

#### Meeting #6:

**Date:** 15/05/2026

**Location:** Online / MS Teams

**Period:** 11:00–12:00 (1.0 Hour)

**Attendees:** Assoc. Prof. Dr. Ömer Korçak, Azra Çetintürk, Ozan Yıldırım

**Objectives:** Verify the mathematical correctness of the global-local consistency regularisation loss.

**Decisions and Notes:** The KL-Divergence implementation was successfully reviewed. The parameters for the upcoming ablation studies (focusing on  $\lambda_0$  and  $\tau$ ) were defined.

#### Meeting #7:

**Date:** 05/06/2026

**Location:** Library Study Room

**Period:** 14:00–16:30 (2.5 Hours)

**Attendees:** Azra Çetintürk, Ozan Yıldırım

**Objectives:** Analyse jury feedback from the PSD defence and structure the Analysis and Design Document (ADD).

**Decisions and Notes:** Responsibilities for updating the Introduction, System Architecture, and Task Plan were divided. Decided to expand the methodology section with detailed pseudocode and system block diagrams as requested by the jury.

### 6.3 Task Plan with Milestones

To address structural complexities, the project has been decomposed into distinct phases and specific subtasks. Responsibilities are clearly divided: Azra Çetintürk leads the server-side aggregation, consistency loss programming, and data augmentation, while Ozan Yıldırım manages the Dirichlet partitioning, client-side training loops, and pseudo-labeling modules.

Table 3: Task/Calendar-Schedule

| Task No | Task Description                       | Expected Output                          | Months (1-12) |
|---------|----------------------------------------|------------------------------------------|---------------|
| P1      | Literature review and method selection | Finalized literature review document     | 1             |
| P2      | Dataset preparation and system setup   | Completed and tested data pipeline       | 2, 3          |
| P3      | Baseline FL implementation (FedAvg)    | Functional FedAvg training loop          | 3, 4          |
| P4      | FedLabel algorithm implementation      | Semi-supervised update phase             | 5, 6, 7       |
| P5      | Implementing the framework             | Modular architecture delivery            | 7, 8, 9       |
| P6      | Evaluation and performance analysis    | Benchmarking charts and ablation studies | 10, 11        |
| P7      | Final documentation and reporting      | Final report and presentation            | 11, 12        |

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